
The Overshoot Deficit: A Cross-Market Conditional Scaling Relation in Directional Change

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Abstract

The Glattfelder–Dupuis–Olsen scaling law, $\langle \omega \rangle \approx \delta$, supplies the zero-profit benchmark for the idealised canonical directional-change round trip: if confirmation occurred exactly at δ , entering at the confirmation point and exiting at the next would return zero. Persistent departures from this zero-profit benchmark are precisely where the trading edge resides.

We make that decomposition explicit on VN30F1M index futures, reporting every result across the full range of plausible execution costs. Profit splits into a discretisation term and an overshoot deficit, and the deficit dominates at practical thresholds. The variable that indexes it is neither θ nor time scale but δ measured against the session's realised range: bucketing on that ratio collapses six thresholds spanning a twentyfold range of θ onto one curve, crossing unity at $\delta \approx 20\%$ of daily range. Because a focal leg can itself contribute to the realised session range used in the denominator, we pre-registered two tests of this self-inclusion — using a disjoint range window and removing each leg's own ticks from the range — and the collapse survives both.

Variation in the Hurst exponent does not explain this: the noise-corrected local exponent is indistinguishable at the horizons of the widest and the narrowest threshold while $\langle \omega \rangle / \delta$ changes twofold. Finite-session spatial compression is the natural remaining explanation, and we tested it rather than asserting it. On a second market and a different feed — broker quotes rather than exchange trades, a distinction we keep throughout — pre-registered and run once on data withheld from every earlier inspection, the relation itself replicates, including a unity crossing at 0.21 to 0.23 against 0.20 here. The compression prediction does not. **The conditional overshoot relation is robust; its mechanism remains unresolved.**

The relation was fitted on data ending 24 April 2026 and then confronted with the 85 sessions that follow, held out from all model development and specification. It predicted $\langle \omega \rangle / \delta = 0.735$ against 0.737 realised, and anticipated that the widest threshold would nearly cease to fire, which it did. The departure it identifies is not marginal. The rule that follows from it reads no profit and loss anywhere in its construction and breaks even only at an execution cost of **1.13 index points per leg**, four times the estimate for an unconditional liquidity taker. Under five-fold anchored walk-forward, with the threshold and holding rule reselected at each fold from preceding data alone, it is **profitable after costs in every fold**, returning a stitched out-of-sample Sharpe of **1.79** at a Pardo efficiency of 80%.

The relation is descriptive and economically relevant, yet it does not become a control input, and we establish why in three steps rather than asserting it. The session range is partly predictable from within the session, reaching a rank correlation of 0.76 by mid-morning. The regime it defines becomes correctly classifiable at 10:30, though not at 09:45. And a pre-registered rule that sets the threshold from that estimate, run once, still loses to a fixed threshold on net profit, Sharpe, drawdown and break-even cost. Observing a state variable well enough to name the regime is not the same as observing it well enough to steer on it. The diagnostic nevertheless carries the economics: across six thresholds, realised $\langle \omega \rangle / \delta$ tracks break-even cost with a rank correlation of -0.943 . Break-even reaches zero at $\langle \omega \rangle / \delta \approx 1.14$ rather than at unity, and the gap is the discretisation cushion, measured.

Keywords: directional change · overshoot · intrinsic time · scaling laws · market microstructure · index futures · walk-forward analysis

1 Introduction

Directional change (DC) replaces physical time with an event clock: an event is recorded when price reverses by a threshold θ from the running extreme. The decomposition splits each price leg into a *directional-change* segment of size $\delta = \theta \cdot P$ and an *overshoot* ω that runs beyond it before the next extreme forms. Glattfelder, Dupuis and Olsen [1] established empirically across foreign-exchange markets that $\langle \omega \rangle \approx \delta$, one of twelve scaling laws they report; Aloud, Tsang, Olsen and Dupuis [2] formalised the event-based framework that surrounds it. The relation is generally reported as unconditional, and the associated claim that overshoot length is insensitive to volatility, scaling with the threshold regardless of the prevailing level, is stated in the same literature. Both are cited widely and used rarely. We think that undersells the first and overstates the second.

Three lines of work follow from that framework. The first carries the scaling description into trading design, most directly in the Alpha Engine [3]. The second takes the description to markets other than foreign exchange: Ma, Xiong, He and Zhang [4] to Chinese equities, and Ao and Tsang [5] to five stock indices including the Hang Seng, where they report an average overshoot length of 0.996 times the threshold — the unconditional law, holding on daily closing prices. The third, and by now the largest, uses directional change as a representation for forecasting and trading: evolutionary strategy search [6], backlash agents [7], regime-change detection [12], machine-learned reversal prediction [8], multi-threshold algorithmic trading across twenty currency pairs [9], and nowcasting the confirmation itself [10]. A recent contribution [11] examines a measurement problem inside the framework, showing that the gap between theoretical and actual confirmation points distorts algorithm performance.

Previous work has therefore examined the unconditional average overshoot law across foreign-exchange markets and, subsequently, across equity indices including the Hang Seng, and has used directional changes primarily for forecasting and trading. **We have not found prior work asking whether the average-overshoot relation is itself conditional on the spatial scale of the threshold relative to the realised price range.** That is the question this paper puts.

The paper begins from an observation we found uncomfortable once we understood it. The most natural DC trade, which enters when a reversal is confirmed and holds until the next confirmation, earns the retrace at confirmation minus the overshoot. If the scaling law held exactly, and if confirmation occurred exactly at δ , the trade would return precisely zero. *The scaling law is therefore not background theory for the strategy; it is the strategy's null hypothesis.* Everything the trade earns is a measurement of departure from it.

Read as a null hypothesis rather than a regularity, the law does real work. It tells you in advance that your P&L is a residual, so the question after "does this make money" becomes "which term of the residual is producing it, and what moves that term". Our central empirical result is that the relation is not unconditional. On this instrument $\langle \omega \rangle / \delta$ ranges from 1.34 to 0.49, a 2.7-fold variation that is systematic rather than noise, and it is indexed by the threshold measured against the session's realised range. We make the decomposition explicit, measure both terms on Vietnamese index futures, and identify what indexes the larger. We then show that identifying the indexing variable does not translate into a better rule, and that the mechanism behind it survives our own attempt to test it no better than it survives the alternatives we rule out.

2 Setup

2.1 Data

VN30F1M is the front-month futures contract on the VN30 index, traded on the Hanoi exchange with a point value of 100,000 VND. Trading is split by a midday break: our sample runs from 09:15 to 14:45 and contains no observations between 11:30 and 13:00, a fact Section 4 makes use of. Our sample is 18,089,838 one-second last-trade observations across 1,340 sessions from 25 February 2021 to 26 August 2026. Prices are forward-filled within each session only; no backward fill is applied at any point, a point we return to in Section 10.

The series is constructed by concatenating the one-second last-trade price series of successive front-month (F1) contracts. It is not a vendor-supplied continuous contract and no back-adjustment is applied. Contract transitions occur between trading sessions. Because all directional-change states, extrema, session ranges and positions are reset at each session boundary, overnight roll gaps do not enter any reported directional-change leg or profit calculation.

As a check, none of the 37 one-second price changes exceeding 15 index points occurred on an expiry date, while the larger discontinuity associated with contract rollover appears in overnight gaps — median 9.80 points after expiry against 4.40 points on ordinary days — outside the analysis window.

2.2 Cost

Round-trip cost is the single input most capable of reversing a verdict in this literature, so we do not assume one. The figure used throughout is **0.175 index points per leg, or 0.35 per round trip**, calibrated from realised execution on this contract, inclusive of commission, spread and latency, rather than inferred from quoted prices. Every result is additionally reported across the interval $[0, 1.00]$ index points per leg, and Table 4 gives that sensitivity in full.

For readers without access to execution records, a Roll [13] estimate on transaction prices with a Corwin–Schultz [14] cross-check gives 0.593 points per round trip. That is the charge appropriate to a participant who crosses the spread on every order and does not manage execution, and it is therefore an upper bound rather than a central estimate: a participant who posts passively and declines to trade when the book is thin pays materially less. Substituting it throughout changes no qualitative conclusion in this paper, because the reversal rule remains profitable well above it.

2.3 External-market data and replication design

The external replication uses Dukascopy HKG.IDX/HKD bid–ask quotes from 25 February 2021 to 26 August 2026. The downloaded archive contains **63,154,002** valid quote records; the acquisition audit found no structurally corrupt files and no missing quote-bearing interval. Unlike the VN30F1M series, which uses last transaction prices, the Hong Kong feed contains quotes rather than trades. We therefore define price as the bid–ask midpoint. This distinction is carried explicitly into the interpretation: the experiment is a cross-market, cross-feed replication and *not* an exchange-level replication.

Raw quotes are converted from UTC to Hong Kong time (UTC+8), sampled onto a one-second grid, and forward-filled only within the relevant trading window; no backward fill is used and no value is carried across a window boundary. The feed is active for approximately 15 of 24 hours. Its complete tradable window runs from 09:00 Hong Kong time to 01:00 on the following calendar day, so observations are grouped by trading window rather than by calendar date.

The replication defines three nested price windows. **Level A** uses the complete tradable window, 09:00–01:00. **Level B** uses the regular Hong Kong day session, 09:15–12:00 and 13:00–16:30. **Level C** treats the morning and afternoon day sessions as separate sub-sessions. Directional-change state and the realised range R are reset at the beginning of

every analysis window; consequently each Level-C leg is normalised by the range of its own sub-session.

No Hong Kong-specific directional-change parameter is introduced. The external analysis uses the same six thresholds, the same δ/R band boundaries, the same minimum requirement of 60 legs per cell, and the same cell-level two-way analysis used for VN30F1M. For each window definition we test whether mean $\langle\omega\rangle/\delta$ declines with δ/R , estimate the interpolated unity crossing, measure the residual contribution of threshold identity after conditioning on the ratio bands, and estimate the compression slope β_C . The window-compression hypothesis predicts

$$\beta_{C, \text{sub-session}} < \beta_{C, \text{day}} < \beta_{C, \text{full}} < 0$$

The hypotheses, band definitions, effect-size criteria and the ordering above were fixed before the governed Hong Kong analysis was executed. The chronology of the interim inspection, the implementation correction and the untouched held-back evaluation is reported separately in Section 8.

2.4 Validation protocol

Directional change is unusually easy to get wrong. The event structure is defined recursively, extremes are identified only in retrospect, and a single backward-looking line produces results that look spectacular and mean nothing. We therefore treat validation as a deliverable rather than a formality: every check below is a script that runs, and every number it produces is stated here so that a reader running the same code can confirm it or contradict it.

V1 Bit-exact reference. The compiled DC implementation is checked against an independent reference implementation across four threshold configurations. **Zero** differing values.

V2 No repainting. The decomposition is recomputed on truncated prefixes of every session. Confirmed extremes, TMV and event times change in **none of 42,000** cases; only the unconfirmed running state moves, which is what should happen.

V3 Streaming equals batch. A tick-by-tick implementation reproduces the batch DC state exactly, with a maximum probability difference of 3×10^{-8} attributable to floating-point ordering. What is backtested is what a live system computes.

Two further checks, **V4** and **V5**, exercise the fitted-model pipeline described in Appendix A: shuffling the targets collapses holdout AUC to 0.4995–0.5002, and planting one tick of future information makes the model outperform the honest version, +369 against +326. The second is the check we consider decisive, because it establishes that the suite has the power to detect what it is testing for. A validation suite that has never been shown to fail is not evidence.

One defect was found and removed during this process: a backward fill affecting 5,540 ticks across two sessions, both inside the specification window. Its effect on any reported number is nil. We removed it because a validation script containing a look-ahead is worth less than none. The tick-level reconciliation performed when the evaluation data arrived is described in Section 5.

3 The overshoot deficit

Let a leg run from the previous extreme to the current extreme, with overshoot ω measured from the directional-change confirmation point to the extreme, and let r denote the realised retrace from the extreme to the next confirmation. Holding a position against the just-confirmed direction over that interval returns

$$\Pi = r - \omega = (r - \delta) + (\delta - \omega) \tag{1}$$

The two terms have entirely different origins. The first, $r - \delta$, is a discretisation effect: confirmation fires on the first tick that crosses the threshold, and price overshoots that crossing by roughly one typical jump. It is always positive and it scales like a constant number of ticks. Fitting $\langle \omega \rangle / \delta = 1 + k / \delta_{\text{ticks}}$ over $\theta \in [5 \times 10^{-4}, 2 \times 10^{-3}]$ gives $k = 3.61$ ticks with residuals of 0.009, 0.059 and 0.043, which accounts for the mid-range almost exactly. Note what this term cannot do. It can never push $\langle \omega \rangle / \delta$ below unity, so it cannot be the source of the effect we are after. Ao and Li [11] treat the same gap between the confirmation the threshold specifies and the confirmation a discrete tape delivers as a distortion to be removed, by substituting theoretical confirmation points. We take the opposite route and measure it, because in this decomposition it is not noise but the first term of the profit, and Section 7 shows it is what separates the crossing of unity from the boundary of profitability.

The second term, $\delta - \omega$, is the *overshoot deficit*. It is the amount by which the market fails to satisfy the scaling law. It changes sign. At $\theta = 10^{-2}$ it supplies 71% of the trade's profit, at $\theta = 5 \times 10^{-3}$ almost none, and at smaller thresholds it is negative and defeats the first term outright.

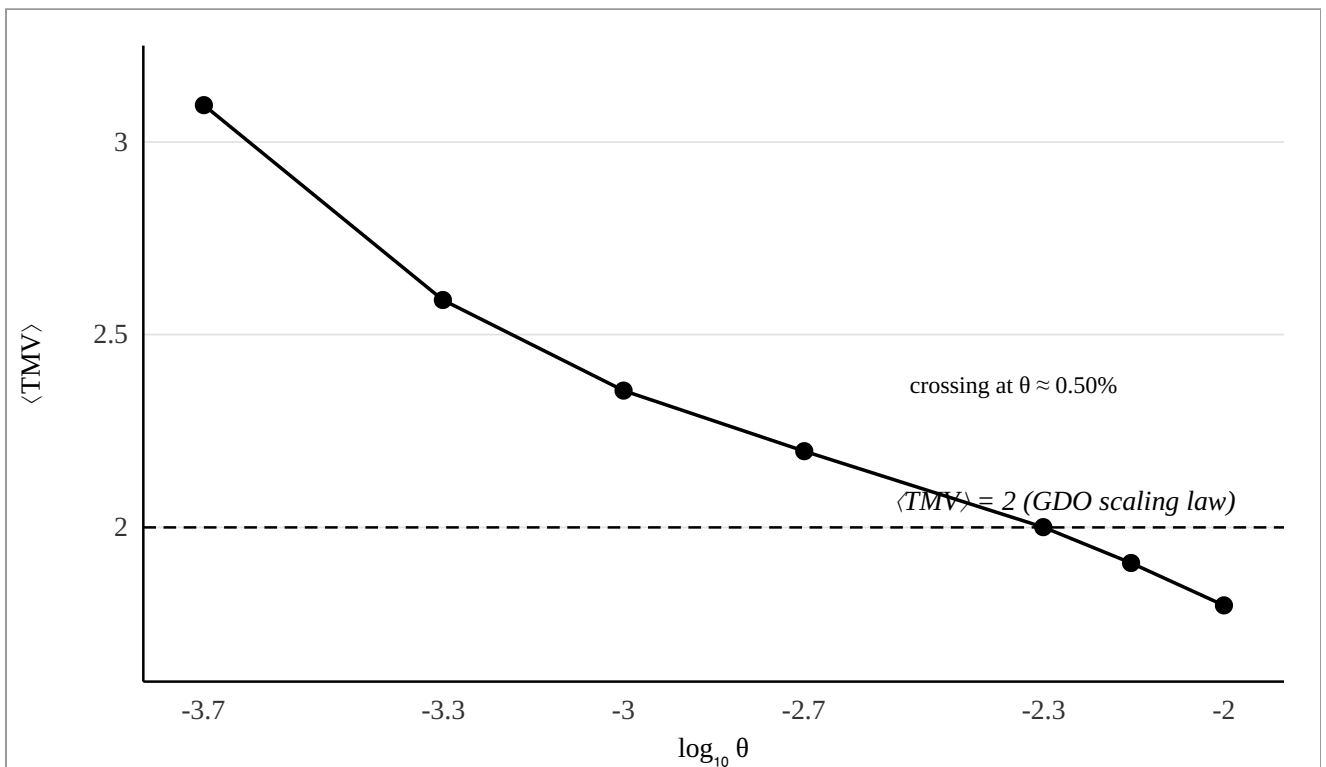


Figure 1. Mean total-movement $\langle \text{TMV} \rangle$ against threshold on VN30F1M. A random walk satisfying the GDO relation gives $\langle \text{TMV} \rangle = 2$. The measured value falls monotonically through that level as θ grows, crossing near $\theta \approx 0.50\%$. Legs above the line have overshoots longer than the threshold (momentum is paid); legs below have overshoots that fall short (reversal is paid).

4 What indexes the deficit

The obvious candidate is time scale. Median leg durations across our thresholds span 55 seconds to 44 minutes, and $\langle \omega \rangle / \delta$ falls monotonically from 1.59 to 0.80 over that range, which invites a Hurst interpretation: trending at short horizons, reverting at long ones.

We find no evidence for it. Raw variance ratios are deflated by microstructure noise in the one-second base interval. Fitting $\text{Var}(r_q) = A \cdot q^b + c$ on $q \geq 20$, a noise-corrected reading of the variance-ratio statistic of Lo and MacKinlay [15], separates the noise floor, $c = 0.0123 \text{ pt}^2$ or 0.078 points per side, from the diffusive component and yields $b = 0.950$, i.e. $H = 0.475$. Local slopes computed after that correction fluctuate between 0.43 and 0.53 about a mean of 0.48, with no monotone drift across three decades (Figure 2). The comparison that decides it is at the two ends. At

the horizon of the narrowest threshold, where $\langle \omega \rangle / \delta = 1.59$, the local exponent is 0.489; at the horizon of the widest, where $\langle \omega \rangle / \delta = 0.80$, it is 0.483. *The same exponent accompanies a twofold change in $\langle \omega \rangle / \delta$* , and at the short end the sign is the wrong way round: H sits below the martingale line where a Hurst account needs it above. The observed scale variation in local H does not account for the overshoot pattern.

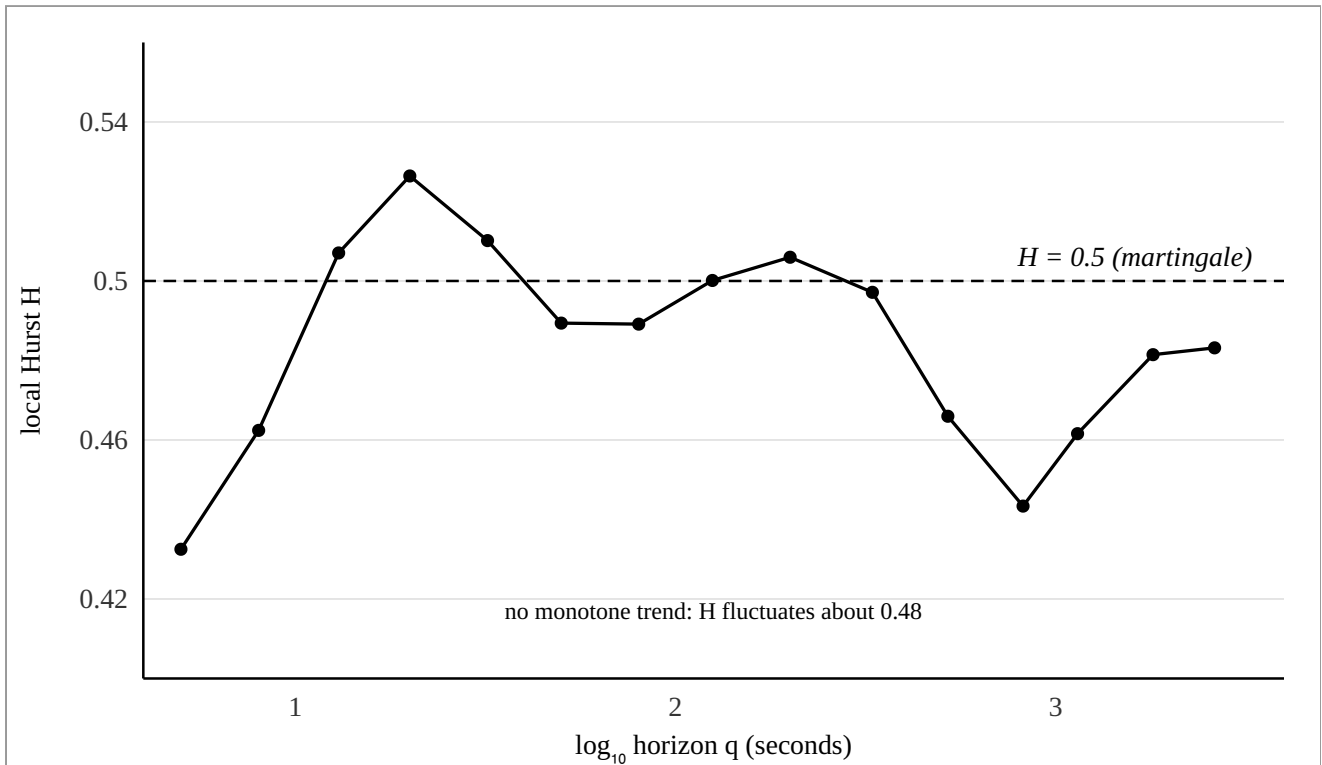


Figure 2. Noise-corrected local Hurst exponent against horizon. Three decades, no monotone drift, and no separation between the horizons of the widest and the narrowest thresholds. Whatever drives the overshoot deficit, it is not scale-dependent memory in the return process.

The variable that does index it is δ measured against the *session's realised range*. Figure 3 is the result we would keep if we could keep only one. Bucketing legs by that ratio and plotting $\langle \omega \rangle / \delta$ within each bucket collapses six thresholds spanning a twentyfold range onto a single curve. In the 0.45–0.65 band, four thresholds give 0.504, 0.517, 0.465 and 0.483. The spread is 0.052, across parameters differing by a factor of five. We searched for an artefact that would explain the collapse away and found none. Six thresholds, a twentyfold range, one curve. We have not seen a cleaner result on this instrument and do not expect to.

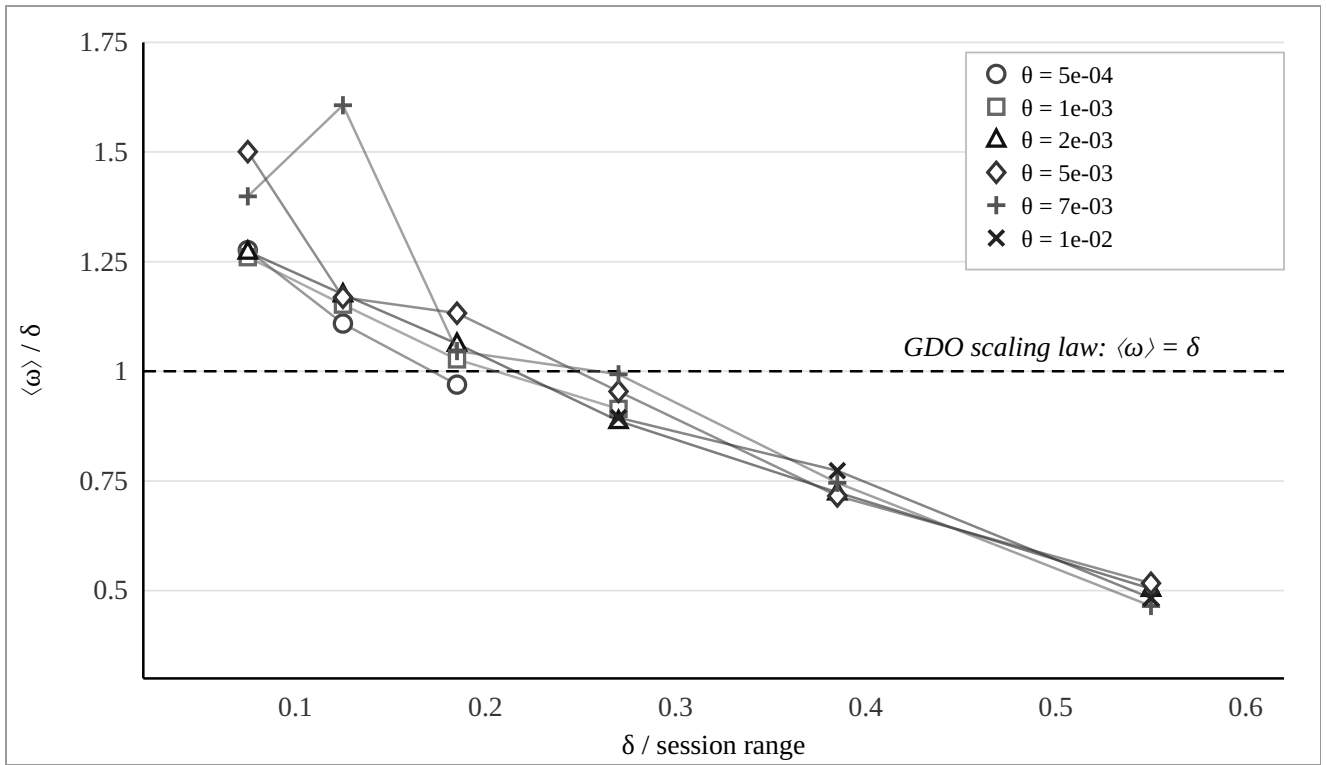


Figure 3. The collapse. $\langle \omega \rangle / \delta$ against δ divided by the session's realised range, with each threshold plotted as a separate series. The curves lie on top of one another: θ is not the indexing variable. The relation crosses unity at $\delta \approx 20\%$ of daily range and falls to 0.5 when the threshold consumes half the day. The candidate mechanism is *spatial compression* — once the threshold has consumed a large fraction of the day's range, less distance remains for the overshoot to run. Section 8 tests that candidate on a second market and does not support it.

The collapse is visual, so we test it. Taking the cell means of Figure 3 as the unit of analysis (28 cells with at least 60 legs each) and fitting a two-way model in band and threshold, adding the δ/R band to a model that already contains θ gives $\mathbf{F}(5,17) = 45.13$, $\mathbf{p} = 3.19 \times 10^{-9}$. Adding θ to a model that already contains the band gives $\mathbf{F}(5,17) = 2.08$, $\mathbf{p} = 0.118$. The band explains 21.7 times the variance that θ does. Threshold has no detectable effect on the curve once the ratio is controlled, which is the collapse stated as a hypothesis test rather than as a picture. The six band means, which a reader can reproduce from the released extract, are 1.342, 1.243, 1.048, 0.929, 0.740 and 0.492 over 745,658 legs. Figure 3 and this test are measured on all 1,340 sessions. The prediction confronted with withheld data in Section 5 is a different object: it was computed from the curve fitted through 24 April 2026 alone, so that test is not circular.

One objection to Figure 3 deserves an answer rather than an assurance. The realised session range contains the leg being measured, so a large overshoot can widen R , lower δ/R , and induce a negative association mechanically. We pre-registered two tests of it before running either.

The first replaces the denominator with the realised range of the *opposite* trading subsession. The exchange's lunch break makes the two windows disjoint, so the measured leg cannot enter its own denominator. The negative association survives at all six thresholds, and analysed separately by subsession the detached correlations retain 84% and 87% of their magnitude. Pooling the subsessions attenuates the relation further because morning and afternoon ranges differ systematically, and the detached coordinate then mixes legs whose true ratios differ; the size of that attenuation is broadly consistent with an errors-in-variables account. This design tests the existence of compression but cannot test the threshold's residual role, because $\delta = \theta \cdot P$ exactly, so a mismeasured denominator hands θ back explanatory power over the residual whether or not that role is real.

The second is surgical. We recompute each leg's denominator over the same session with that leg's own ticks removed. Same session, same volatility, no proxy, and the only path cut is the direct one from ω_i into R_i . Its useful property is that most legs never reach the session high or low, so their denominator does not change at all: **2.7% of legs are affected**, and those are precisely the large-overshoot legs, averaging $\langle\omega\rangle/\delta = 2.21$ against 1.47 for the rest.

The result splits cleanly, and we report both halves. **The threshold's residual role is unaffected and in fact measures smaller**: p_0 rises from 0.118 to **0.555** and the variance ratio falls from 0.046 to 0.036. Removing self-inclusion does not weaken the collapse, which also explains why the detached design appeared to contradict it. **The steepness, however, is overstated by about a sixth**: the compression slope moves from -1.768 to **-1.454** , a flattening of 17.8%, concentrated exactly where the mechanism predicts — the deepest band moves from 0.492 to 0.633 while the shallowest moves only from 1.342 to 1.333. The corrected relation still spans 1.333 to 0.633 and still crosses unity between the same two bands.

Stated compactly: removing the focal leg from the session range reduces the estimated compression slope from -1.768 to -1.454 , a 17.8% attenuation, while *strengthening* rather than weakening the threshold's residual role, and the unity crossing is unchanged. Mechanical self-inclusion therefore exaggerates the magnitude of compression but does not generate the collapse. We quote the measured slope throughout and state this correction rather than folding it into the error bars. The figure is a directly measured correction for this specific intervention, not a claim to have identified every source of bias that might sit in Figure 3; what it rules out is the one objection that would have made the collapse an artefact of its own construction.

This matters less for the slope than for what the paper actually rests on. Nothing here depends on the compression slope being -1.768 rather than -1.454 . What has to survive is that $\langle\omega\rangle/\delta$ falls as δ/R rises, and that the fall carries the ratio across unity, because unity is where the overshoot-deficit term changes sign and the trade stops being paid for momentum. Both survive the correction intact.

The agreement tightens where it matters. Across thresholds within a band, the spread of $\langle\omega\rangle/\delta$ is 0.052 and 0.057 in the two highest ratio bands, against a range of 0.850 between bands. In the low-ratio bands the spread is wider, 0.24 and 0.50, the latter driven by a single cell ($\theta = 7 \times 10^{-3}$ at ratio 0.10–0.15) that is visible as the outlier in Figure 3. That region is where $\langle\omega\rangle/\delta$ exceeds unity and the reversal trade is not taken, so the dispersion there does not bear on the result.

5 Out-of-sample validation

The evaluation window is temporal, not random: every session after 24 April 2026. The curve, its bucket boundaries and the threshold set were fixed on the specification window and were not revised afterwards. The ordering is stronger than a convention about not looking. The specification scripts were written and executed before the file containing the later sessions was present on the machine at all, so the evaluation data could not have influenced the relation by any route, deliberate or otherwise. The overlapping period of that file was then verified tick by tick against the existing panel before any result was computed: 1,012,575 matched observations, maximum absolute price difference 0.000000.

The curve fitted on data through 24 April 2026 predicts $\langle\omega\rangle/\delta = 0.735$ for the ratio band the withheld window actually occupied. The realised value is **0.737**, an absolute error of 0.003, or 0.39% of the realised value. Predictions this close are usually luck and we would normally say so. What makes us treat this one differently is the second prediction: the same curve said the widest threshold would stop firing, and it did, in a way that costs us trades rather than flattering the result. The curve also predicted the unfavourable consequence correctly: at $\theta = 1\%$ the threshold $\delta \approx 19.7$ points against a median daily range of 24.1 points, so the rule should almost cease to fire. It produced **seven legs in eighty-five sessions**.

Table 1. Frozen configurations on the 85 withheld sessions. Criteria were registered before the data was seen. Gross per turnover unit is quoted because it is measured per trade rather than per session.

configuration	OOS gross/turn	IS gross/turn	OOS net (pts)	verdict
Classifier + regression (M+B)	-0.217	0.817	-25.1	fail
Confirmation + threshold (A+M)	0.694	0.534	+216.6	pass
DC reversal, $\theta = 0.5\%$	1.448	0.399	+298.0	pass
DC reversal, $\theta = 0.7\%$	3.117	1.130	+258.9	pass

Table 1 separates two distinct outcomes. The failing configuration does not merely fall below cost — its gross turns *negative*. The edge inverted rather than eroded. The two fitted-model configurations are described in Appendix A; the two directional-change configurations, which carry the argument of this paper, use no fitted model at all.

6 Results

Table 2. The directional-change constructions, ordered by Sharpe on the specification window of 819 sessions (2 January 2023 to 24 April 2026). None reads profit and loss in its construction. *Break-even* is the execution cost per leg at which net profit reaches zero, a derived quantity rather than an assumption. The final two columns report the 85-session evaluation window, which is disjoint from the specification window and was not examined during construction. A fitted classifier and its blend with these rules are reported in Appendix A.

construction	trades/day	Sharpe	net	Calmar	break-even	Sharpe*	net*
Reversal, fixed $\theta = 0.7\%$	0.81	2.17	+1,274	6.3	1.13	6.79	+259
Reversal, θ by (TMV) criterion	0.67	1.95	+1,132	5.6	1.20	6.79	+259
Reversal, fixed $\theta = 1.0\%$	0.26	1.87	+851	8.2	2.14	3.56	+50

* evaluation window, 85 sessions.

Table 2 separates two windows that must not be mixed. All constructions were specified on the 819 sessions ending 24 April 2026; the 85 sessions after that date form a disjoint window that was not examined during construction. The reversal rule turns over less than one round trip a day and its Sharpe on the specification window, 2.17, is unremarkable. That is not where its interest lies.

We also ran the reversal rule alongside an independently fitted classifier built on the same event structure. The two are negatively correlated, -0.215 within the specification window and -0.331 on the evaluation window, and blending them improves portfolio performance materially. That construction and its walk-forward are reported in Appendix A rather than here, because it depends on a fitted model whose architecture is not the subject of this paper, and because every claim we make about the scaling relation stands without it.

The break-even column is where it lies, because it is derived rather than assumed: it is the execution cost at which the rule returns exactly zero. The reversal constructions break even between **1.13 and 2.14** index points per leg. Every construction we built on a fitted classifier breaks even between 0.41 and 0.53, a threefold gap that holds whatever cost figure a reader considers realistic. A rule that survives an execution cost four times a liquidity taker's is a different object from one that needs favourable execution to exist, and the difference does not depend on which cost estimate is believed.

Those are the operative figures for a rule whose parameters are not refitted annually. Walk-forward asks a different question, and we answer it separately. The protocol requires a candidate set, so we define one: thirty cells, five thresholds by two entry windows by three exit rules. Each fold selects from that set using only data preceding the test year and is scored on that year alone, so selection is part of what is being tested rather than a step performed before testing. Under this protocol the rule is charged for parameter instability in addition to everything else, and the stitched out-of-sample Sharpe is **1.79** at a Pardo efficiency of **80%**, with every fold positive.

Table 3. Anchored walk-forward on the reversal rule, fold by fold, with selection by Calmar. Each fold picks its cell from a grid of thirty — five thresholds $\times$ two entry windows $\times$ three exit rules — using all data preceding the test year, and is scored on that year alone. WFE is Pardo's [16] walk-forward efficiency, out-of-sample net per session divided by in-sample net per session. The cell marked *thin* was selected on training data but barely fired out of sample. Cell notation: θ | entry window | exit rule, where 09:45+ admits entries after 09:45 only, *full* uses the whole session, and $\leq 120m$ closes the position after at most 120 minutes. The classifier and blended arms are in Appendix A.

fold	train	selected cell	IS Sharpe	OOS Sharpe	OOS net	MDD	WFE
2023	436	1.0% 09:45+ $\leq 120m$	3.91	+3.99	+281	29	62%
2024	681	1.0% 09:45+ $\leq 120m$	3.83	+1.94	+61	13	15%
2025	931	1.0% 09:45+ $\leq 120m$	3.39	+1.32	+297	100	97%
2026H1	1180	1.2% full $\leq 120m$	2.09	+3.97	+163	10	212%
2026H2	1255	1.2% full $\leq 120m$ <i>thin</i>	2.21	+0.00	+61	0	65%
stitched	—	2 distinct cells across 5 folds	—	1.79	+863	100	80%

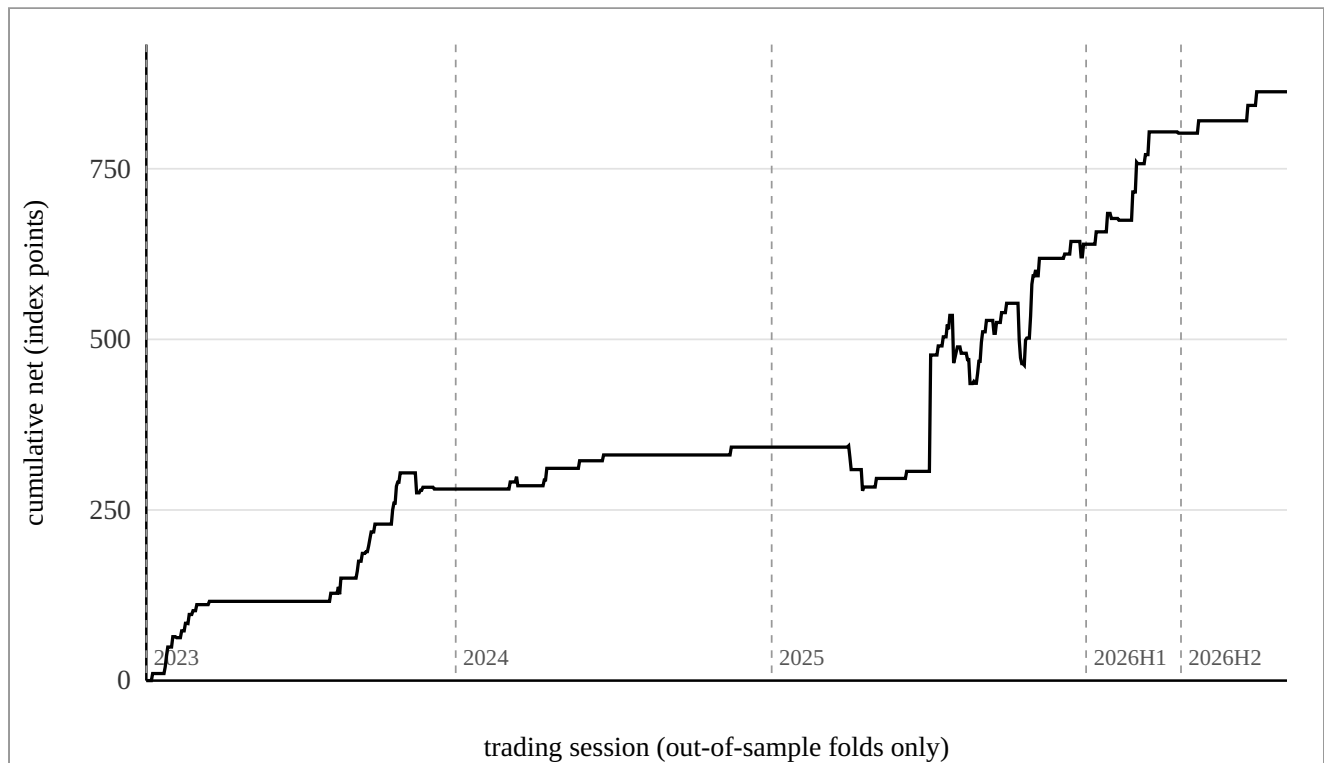


Figure 4. Stitched out-of-sample equity for the reversal rule, comprising only test folds. Vertical rules mark fold boundaries. Every fold is positive; the weakest, 2024, is where the method is genuinely tested, and the strongest are periods in which the entire parameter grid improved.

The aggregate efficiency of 80% clears Pardo's 50% threshold, but per-fold efficiency is where the honest cost of annual reselection shows: the rule spans 15% to 212%, and its weakest fold, 2024, retains under a sixth of its in-sample rate while still finishing positive. Two distinct cells were selected across five folds, which is the kind of stability we would want and did not design for. We report the range rather than the aggregate because a reader deciding whether to trade this would care about the bad fold, not the average one.

Table 4 gives the cost sensitivity promised in Section 2.2. The reversal rule, which fits nothing, is still positive at 1.00 index points per leg. Every construction in this study that uses a fitted model turns negative somewhere between 0.40 and 0.60. Whatever a reader believes about execution quality on this contract, that ordering does not change, and it is the reason the rest of this paper is about the rule that fits nothing.

Table 4. Net profit in index points over the combined specification and evaluation history, as a function of cost per leg. Column 0.2965 is the Roll/Corwin–Schultz taker estimate; 0.175 is the measured execution benchmark of Section 2.2. The reversal rule remains profitable at 1.00 per leg, roughly three and a half times the taker figure. Constructions that use the fitted classifier turn negative between 0.40 and 0.60 and are tabulated in Appendix A.

cost per leg →	0	0.0875	0.175	0.2965	0.4	0.6	1.00
Reversal rule, $\theta = 0.7\%$	+1,781	+1,657	+1,533	+1,360	+1,213	+928	+359
Reversal, causal θ	+1,599	+1,495	+1,391	+1,247	+1,124	+886	+411

7 An information constraint on the scaling relation

Section 4 identifies the indexing variable, which suggests an obvious improvement: set the threshold each day so that δ is a fixed fraction of the session's range. Estimating that range causally from the trailing twenty-session median, this *underperforms a fixed threshold on every metric* — Sharpe falls to 2.00 from 2.59, gross per turn to 0.657 from 1.242, and maximum drawdown roughly doubles.

That result is easy to over-read. It fuses two quite different failures: the range may be unpredictable, or it may be perfectly predictable and still useless as a control input. We separated them. The answer is the second, and it is more interesting than the first.

The question is adjacent to, but not the same as, nowcasting a directional change before its confirmation [10]. That literature asks whether the event can be seen coming. We ask whether the state variable the event lives in can be estimated well enough to set the threshold that defines the event, which is a control problem rather than a prediction problem, and the two have different answers.

The range is partly predictable. Refitting before every session on strictly prior sessions only, a log-regression of the full-session range on the elapsed range and the trailing twenty-session median reaches a rank correlation of 0.695 and an out-of-sample R^2 in logs of 0.460 by 09:45, and 0.759 and 0.563 by 10:30, on the 1,200 sessions that carry a causal estimate at both times. The adaptive rule that failed was working with about half the information available to it.

Prediction is nevertheless the wrong target. A threshold rule does not need the range; it needs to know which side of the unity crossing the session will fall on. That is a classification problem, and an easier one. Binning legs by the *estimated* ratio rather than the realised one, and counting only legs entered after the estimate is formed, the picture changes sharply with the hour. At 09:45 the estimated bands are not even ordered correctly, and the bucket the estimate labels as the fade zone realises $\langle \omega \rangle / \delta = 1.059$ — the wrong side of unity, so the label is worse than useless. At 10:30 the estimated bands are monotone and that same bucket realises 0.937, capturing 80.6% of the distance between no conditioning at all (1.492) and the ceiling set by ex-post labels (0.803). The state variable becomes readable roughly seventy-five minutes into the session, and not before.

Better observation does not produce better control. We pre-registered the resulting rule before computing anything: decision time 10:30, $\delta^* = 0.32 \cdot \hat{R}$ with $\hat{R}$ from the model above, $\theta^* = \delta^* / P$, state reset at the decision time, trading only afterwards, one-second execution delay, and the identical treatment applied to fixed-threshold comparators. It was run once. The adaptive rule loses on every axis, including the mechanism diagnostic it was built to exploit (Table 5). It remains profitable in each of the six calendar years and beats four of the six fixed thresholds; it simply loses to the two a constant choice would have found. Because this evaluation uses the full history rather than the withheld window, we report it as a locked historical evaluation and not as an out-of-sample test.

Table 5. Causal threshold control against fixed thresholds, 1,200 sessions. All arms reset their directional-change state at 10:30 and trade only afterwards, so the comparison isolates the choice of threshold. The adaptive arm sets θ from an estimate formed at 10:30 from prior sessions only; the fixed arms are granted hindsight in choosing their own threshold. *Break-even* is the execution cost per leg at which net profit reaches zero.

threshold rule	net (pts)	Sharpe	max DD	break-even	round trips	$\langle\omega\rangle/\delta$
Adaptive, $\theta^* = 0.32 \cdot \hat{R}/P$ at 10:30	+1,194	1.53	189	0.483	1,940	0.936
Fixed, $\theta = 0.7\%$	+1,583	2.35	104	1.110	847	0.837
Fixed, $\theta = 1.0\%$	+1,290	2.27	70	2.311	302	0.717
Fixed, $\theta = 0.5\%$	+708	0.90	379	0.363	1,882	0.978
Fixed, $\theta = 0.2\%$	-7,451	-5.55	7,455	-0.104	13,360	1.184
Fixed, $\theta = 0.1\%$	-28,181	-11.28	28,185	-0.156	42,574	1.359
Fixed, $\theta = 0.05\%$	-71,839	-15.14	71,828	-0.143	112,863	1.598

The constraint is therefore sharper than the one we originally stated. It is not that the state variable is unobservable during the session — by mid-morning it is largely observable. It is that observing it well enough to classify the regime is not the same as observing it well enough to steer a threshold, and the gap between those two standards is where the benefit dies.

The relation nonetheless carries economic content. Read Table 5 down its last two columns. Across six thresholds spanning a twentyfold range, the realised $\langle\omega\rangle/\delta$ predicts the break-even cost with a rank correlation of -0.943 ($p = 0.005$). Unity is the zero of the overshoot-deficit term, not of the trade: by equation (1) the discretisation term $r - \delta$ stays positive there, so at $\langle\omega\rangle/\delta = 1$ the rule still breaks even at $+0.31$ points per leg, and break-even reaches zero only at $\langle\omega\rangle/\delta \approx 1.14$. That gap is the discretisation cushion of Section 3 expressed in cost, and it is the reason the two crossings do not coincide. The scaling departure is not a descriptive statistic that happens to accompany a profitable rule: it orders the profitability of six independently constructed rules that were never fitted to each other. What it does not do is supply a steering input.

The failure of the adaptive rule also cautions against assuming that normalisation by a contemporaneous scale estimate necessarily improves a fixed-threshold decision rule. Our prior on adaptive thresholds is now low enough that we would want a strong argument before testing another.

8 External replication on a second market

Section 4 identifies a candidate mechanism by elimination. Elimination is not evidence, so we pre-registered a direct test of it, hashed each document and anchored it in the Bitcoin blockchain before the work it governs was run, and executed the analysis once.

The instrument is a Hong Kong index feed over the identical window, 25 February 2021 to 26 August 2026, giving 63,154,002 valid quotes, with no missing quote-bearing interval. The design varies the price window while holding instrument, feed and tick generation constant: the *full tradable window*, the *regular day session*, and the *morning and afternoon sub-sessions* taken separately. If finite available price space drives the relation, compression should strengthen monotonically as the window narrows.

Chronology, disclosed. Roughly 60% of the sample was inspected while acquisition was still running, first under an implementation later found to group observations by calendar date rather than by the tradable window, which crosses midnight, and again after that defect was corrected. The correction was registered, hashed and timestamped before any corrected statistic existed. Everything after 26 June 2024 was never examined until the single execution reported here and is reported first. The full-sample figures are therefore a pre-registered analysis with an interim inspection of approximately 60% of the sample, not a blinded confirmatory result. The held-back period carries no such qualification.

Table 6. Pre-registered external test, held-back period only: 27 June 2024 to 26 August 2026, never examined before this execution. Criteria were fixed in advance and were not altered after the interim inspection. *Crossing* is the interpolated δ/R at which $\langle\omega\rangle/\delta = 1$, against a registered window of [0.15, 0.30] and a discovery-market value near 0.20.

price window	legs	cells	band effect	crossing	β_C
full tradable window	291,255	24	$F(5,13) = 244.8$	0.218	-1.391
regular day session	195,442	20	$F(5,10) = 37.7$	0.226	-1.534
sub-sessions	192,676	23	$F(5,13) = 82.0$	0.210	-1.372

The relation replicates. Monotonicity was present under all three window definitions in the untouched holdout and remained present in the full-sample estimates, with band effects between $F = 37.7$ and $F = 244.8$. The two layers are not independent — the full sample contains the holdout — so the holdout carries the evidential weight and the full sample confirms nothing beyond it. What we report is therefore a *cross-market conditional scaling relation indexed by δ/R* , and not a phenomenon of one contract.

The Hang Seng has been examined in this framework before. Ao and Tsang [5] report an average overshoot length of 0.996 times the threshold on daily closing indices — the unconditional law, holding almost exactly. Our measurement does not contradict theirs. It asks a different question of the same market: conditioned on δ/R , the same ratio runs from 1.15 to 0.50, and an unconditional average taken across that range would land near unity by construction. An average that holds is not evidence that the quantity being averaged is constant.

The crossing is the striking result. In the untouched Hong Kong holdout the estimated unity crossings were **0.218**, **0.226** and **0.210** across the three window definitions, against approximately **0.20** on VN30F1M. Despite differences in instrument construction, price level, currency, trading population and observation type, the economic boundary occurs at a remarkably similar fraction of the realised price range. We did not expect agreement this close and we report it without embellishment: the registered acceptance window was [0.15, 0.30], and five of the six Hong Kong estimates fall between 0.21 and 0.23.

The threshold retains a small residual role. The significance condition on θ fails at every window definition in both layers, with p_0 below 0.001 in three cases and between 0.006 and 0.033 in the others. All three pre-registered magnitude conditions pass everywhere: the band effect is significant at $p < 0.01$ in every case; the θ sum of squares is between 3.6% and 15.0% of the band sum of squares against a 35% ceiling; and the within-band spread is between 19.4% and 29.8% of the between-band range against a 40% ceiling. The registered formulation is the accurate one: **conditioning on δ/R absorbs most, but not all, of the threshold dependence.** With two to three times the leg count of the discovery market, a small residual effect becomes detectable; that is why the magnitude conditions were written into the protocol before any of this was measured.

The mechanism prediction fails. Compression was predicted to strengthen monotonically as the window narrowed. It does not.

Table 7. Compression slope by price window. The prediction registered in advance was $\beta_C(\text{sub-session}) < \beta_C(\text{day}) < \beta_C(\text{full window})$. Monotonicity is what was predicted and monotonicity is what fails, in both layers and in the same direction.

layer	full tradable window	regular day session	sub-sessions	predicted order
held-back period	-1.391	-1.534	-1.372	not obtained
full sample	-1.424	-1.510	-1.488	not obtained

Compression does strengthen from the full tradable window to the regular day session, which is the direction the account predicts. But it weakens again at the sub-session level, and that reversal replicated in the untouched holdout rather than appearing only in the portion we had already seen. A prediction that was specific enough to be wrong was wrong.

The reversal at the sub-session level may reflect the simultaneous increase in boundary truncation and in directional-change state resets when windows are shortened, both of which were named as limitations in the protocol before the analysis they govern was executed and before any Hong Kong outcome was inspected. Distinguishing those effects from available price space requires a separate design and is left for future work. We are not going to test additional explanations against the same data until one of them survives.

Limitation. The Hong Kong observations are broker bid–ask quotes represented by their midpoint, not exchange transaction prices. The result therefore constitutes cross-market, cross-feed replication, and not exchange-level replication. That restriction was written into the protocol before any result existed, and it applies to the favourable findings above exactly as it applies to the unfavourable one.

Where this leaves the mechanism. The conditional overshoot relation is robust, but its mechanism remains unresolved. A pre-registered window-compression test did not support the predicted monotonic strengthening of compression as the available price window narrowed. Although compression strengthened from the full tradable window to the regular day session, it weakened again at the sub-session level, a pattern that replicated in the untouched holdout period. Finite-session spatial compression therefore remains a plausible contributor, but cannot account for the observed relation on its own.

9 What does not work

Section 4 finds no evidence for a Hurst-based account, while Sections 7 and 8 show that causal normalisation and the pre-registered spatial-compression mechanism do not explain the relation. This section records additional alternatives that we tested and that did not improve the explanation. These are negative results rather than exhaustive exclusions; the mechanism remains unresolved.

Volume, five independent routes. Five distinct hypotheses about volume, each with a different mechanism and a different test. As additional model features, twenty-three OHLCV columns moved holdout AUC from 0.8088 to 0.8081 and were redundant with an existing magnitude regressor ($p = 0.636$). As a per-leg Kyle-lambda, $|TMV|$ divided by absolute flow was beaten by $1/T$ alone (0.1204 against 0.1019), with residual correlation of 1.3 standard errors after removing the duration component. Volume asymmetry around the extreme changed sign between flow and volume denominators. Flow intensity did survive both a partial-correlation test at 13.1 standard errors and an economic floor, but as a causal gate on the reversal rule it was indistinguishable from a random gate at matched retention ($p = 0.14\text{--}0.87$), and every gated variant had lower net than no gate. On this instrument, at the DC-leg level, every volume aggregate reduces to duration: the information sits in *when* rather than *how much*. Within the leg-level aggregates tested here, volume adds little beyond duration, and we no longer test it on this instrument.

Others. A Hurst account of the deficit (Section 4); adaptive thresholds (Section 7); a nested multi-threshold alignment filter; classical oscillator constructions applied to the model's probability output, which occupy the bottom of every ranking we produce.

10 Measurement pitfalls

Three traps produced spurious results during this work. Each is invisible in summary statistics and each survives ordinary code review. They are documented here because they are the reason the results above can be relied on.

Position-grammar defect. Our execution layer treats a signal of zero as *hold* rather than *flat*, forward-filling the previous position. Because 19.3% of legs are dropped by causality filters, this carried stale positions across the gaps. An early version of the reversal backtest reported a Sharpe ratio of 3.40. An audit showed that **37.3% of its gross**

came from ticks belonging to no trade at all, carried there by the forward fill. With a correct flat-between grammar every momentum variant dies and only the reversal survives. For event-segment strategies, position arrays should be constructed explicitly, with an assertion that exposure equals signal coverage.

Unit substitution in a ratio. The compression ratio computed per session is 0.82; computed per leg it is 0.34. Narrow-range days produce no legs at all, so the leg sample self-selects onto wide-range days. A monitor built on the session-level statistic would have been indexed against a curve built on the leg-level one.

Cost model as a verdict switch. The same reversal rule reads as "at breakeven" under a market-taker cost estimate and as clearly profitable under the measured execution cost. No line of strategy code differs between the two readings.

11 Conclusion

The GDO scaling law fixes the zero-profit benchmark for the canonical directional-change round trip. On VN30F1M the market departs from that benchmark systematically, and the departure is indexed by the threshold's size relative to the session's range rather than by the threshold itself or by scale-dependent memory: six thresholds spanning a twentyfold range collapse onto one curve in that coordinate. The relation predicted a withheld window to within 0.003, including which configuration would nearly cease to be tradeable. That departure is not a curiosity about the law. Measured after cost and under walk-forward, it is the edge.

The rule that falls out of this selects its threshold by a criterion that never reads profit and loss, and it breaks even at 1.13 index points per leg. No construction we built on a fitted classifier reaches half that figure, and that property is what makes the rule deployable. Under walk-forward, with the threshold and holding rule reselected annually from preceding data alone, it returns a stitched out-of-sample Sharpe of 1.79 at a Pardo efficiency of 80%, positive in every fold. Those are smaller numbers than a blended construction produces, and we report them because they belong to the object this paper is about.

The relation is not confined to the instrument it was found on. A pre-registered test on a second market and a different feed reproduces it at every window definition, and puts the unity crossing at 0.21 to 0.23 against 0.20 here, on a market at approximately 16.5 times the price level with a different tick size, currency and participant population. What that test does not reproduce is our explanation of it. Compression was predicted to strengthen monotonically as the available price window narrowed; it strengthens from the full tradable window to the day session and then weakens again, and the reversal replicated in a holdout we had never looked at. We report a cross-market conditional scaling relation indexed by δ/R whose mechanism is unresolved, and we would rather say that than defend a mechanism our own test declined to support.

The relation is descriptive, and the rule derived from it trades, but the relation cannot be used as a control input — and the reason is not the one we expected. The state variable is not hidden until the close. By mid-morning it is estimated well enough to classify the regime correctly, and we show the hour at which that becomes true. A pre-registered rule built on that estimate still loses to a constant threshold on every metric we registered. The binding constraint is the gap between classifying a regime and steering on it, and that gap is measurable rather than rhetorical. We regard this as an information constraint rather than a shortfall, and we suspect it is more common than the literature admits: a fair number of published mechanisms are likely unexploitable for exactly this reason, without anyone having checked. What the relation does supply is a kill condition, and now also a price. Realised $\langle\omega\rangle/\delta$ tracks break-even cost across six thresholds at a rank correlation of -0.943 . The overshoot-deficit term changes sign at unity, while total break-even cost reaches zero only near $\langle\omega\rangle/\delta \approx 1.14$, because of the discretisation cushion. We know the ratio at which this edge stops working, we can measure that ratio daily, and we know how far the market must move to reach it.

12 Data and code availability

A public repository carries the manuscript, the directional-change implementation, the analysis code for every result reported here, all pre-registration documents with their timestamp proofs, and a leg-level extract sufficient to rebuild Figure 3, the two-way analysis, the compression slope and the unity crossing from scratch in a few seconds.

What the extract contains, and why. It reports each leg as dimensionless quantities only: threshold, session date, direction, δ/R , ω/δ , r/δ and duration. It does *not* report δ , ω , r or the session range in index points. That is deliberate rather than incidental. Because $\delta = \theta \cdot P$, publishing δ alongside θ would permit reconstruction of the underlying price at every directional-change extreme, at roughly one-minute resolution across the whole sample — a derived form of licensed tick data we are not free to redistribute. The dimensionless extract reproduces every claim this paper makes about the scaling relation and reconstructs no price.

Economic results. Profit figures cannot be recomputed from that extract, for the same reason, so they are published instead as frozen aggregates: per-construction statistics, the walk-forward folds and their stitched result, the cost-sensitivity grid, the causal threshold-control arms, and the out-of-sample equity series plotted in Figure 4. Each of these is a profit-and-loss difference and none permits recovery of a price level. Raw VN30F1M tick data, and any price-reconstructable derivative of it, are available under a data-access arrangement consistent with the underlying licence rather than by public download.

The Hong Kong arm carries no such restriction. Its data source is free and public, so nothing there is withheld. The leg extract underlying Section 8 is released in full, with a script that rebuilds Tables 6 and 7 from it in seconds; and for a reader who wants leg construction checked as well, the acquisition script is included and reproduces the analysis end to end from the quotes.

Code and replication materials. The manuscript, the analysis code, the pre-registration records with their timestamp proofs, and the released replication extracts are available at <https://github.com/son-d0/overshoot-deficit>, release v1.0. An archived snapshot of that release is available through Zenodo at 10.5281/zenodo.22227011. The repository documents which results are directly reproducible from the released data and which require the licensed VN30F1M tick panel.

We make this available because the claims here are of a kind that should not be taken on trust. The collapse in Figure 3 is either present in the data or it is not; the prediction in Section 5 either lands within 0.003 of the realised value or it does not. Both are cheap to check and we would rather they were checked.

AI use disclosure

Generative AI tools, including Claude Opus 5 (Anthropic), used through Claude Code, and OpenAI ChatGPT (GPT-5.6 family; model configurations varied across sessions), were used for language editing, drafting and implementing analysis code and pre-registration materials, methodological exploration, literature organisation, and quality-control checks. The author independently reviewed and verified the originality and accuracy of the manuscript, the analyses, the reported results and the references; confirmed that the tools were used under terms suitable for this work; determined the final research design and the pre-registered decision criteria; and takes full responsibility for the integrity of the manuscript.

Appendix A The fitted classifier and the blended construction

This appendix records the fitted-model arm of the study. It is here rather than in the main text because the paper's subject is the scaling relation and the parameter-free rule that follows from it, and because every claim about that relation stands whether or not this arm exists. It is reported rather than dropped because it produced the largest

economic figures in the study and omitting it would flatter the main text by selection.

The model. A gradient-boosted classifier is fitted on directional-change events. The unit of observation is a confirmed leg; the label is whether the leg's remaining travel exceeds a threshold; the features are the event-level quantities of Section 3 — total movement, overshoot, duration, and their recent history. The fitted sleeve additionally uses contemporaneous event-window covariates, including aggregated order-flow measures. **These variables are used only by the ancillary classifier reported in this appendix and do not enter the directional-change relation, the reversal rule, or the Hong Kong replication.** The model's output, written p throughout, is the predicted probability for the leg in progress, available at the confirmation tick and not before. It is fitted on the specification window only.

V4 Label permutation. Shuffling the targets five times collapses holdout AUC to **0.4995–0.5002** and net profit to between -40 and -130 points. The pipeline extracts nothing from noise.

V5 Planted leak. Granting the model one tick of future information makes it outperform the honest version, **+369 against +326**. This is the check we consider decisive, because it establishes that V1 through V4 have the power to detect what they are testing for.

Table A1. Constructions driven by the fitted probability p , and blends of those with the reversal rule, on the same two windows as Table 2. Compare the break-even column with the 1.13 to 2.14 of the parameter-free rules.

construction	source	trades/day	Sharpe	net	Calmar	break-even	Sharpe*	net*
Blend, unit weight	$p +$ DC	3.70	3.50	+3,669	16.9	0.78	5.05	+451
Blend, half weight	$p +$ DC	3.16	3.50	+2,607	26.4	0.68	3.41	+296
Confirmation + threshold	p only	2.62	2.34	+1,544	14.7	0.53	1.59	+142
Confirmation + threshold + regression	p only	2.51	2.23	+1,471	14.0	0.53	1.79	+156
Time confirmation, 15 min	p only	2.74	2.15	+1,440	9.4	0.50	1.59	+142
Confirmation + regression	p only	2.51	2.08	+1,382	9.1	0.51	1.64	+143
Time confirmation, 10 min	p only	3.58	1.98	+1,397	10.5	0.41	2.56	+217

* evaluation window, 85 sessions.

Held at fixed parameters the blend returns a Sharpe of 3.50 on the specification window and 3.41 on the evaluation window. The gain is not signal quality: the two sleeves correlate -0.215 in sample and -0.331 out of sample, and the entire improvement is diversification between a fitted model and a rule with no parameters at all.

Table A2. Anchored walk-forward, all three arms, selection by Calmar. The candidate set is 135 classifier cells, 30 reversal cells, and their combinations under six blend weights. The reversal arm is reproduced from Table 3 for comparison.

arm	stitched OOS Sharpe	net	max DD	WFE	distinct cells
Classifier alone	1.66	+1,294	135	108%	3
Reversal alone	1.79	+863	100	80%	2
Blend	2.27	+2,373	224	94%	5

The blend beats both sleeves on net in every fold, but not on drawdown: its stitched maximum drawdown of 224 points exceeds the classifier's 135, and the excess sits in the 2025 fold. All three aggregate efficiencies clear Pardo's 50% threshold. Per-fold efficiency is where annual reselection is paid for: the blend spans 29% to 154%.

Table A3. Cost sensitivity of the fitted-model constructions, over the combined specification and evaluation history, completing Table 4.

cost per leg →	0	0.0875	0.175	0.2965	0.4	0.6	1.00
Blend (unit weight)	+5,276	+4,698	+4,120	+3,317	+2,633	+1,312	-1,330
Blend (half weight)	+3,896	+3,399	+2,903	+2,214	+1,627	+492	-1,776
Classifier (A+M)	+2,515	+2,101	+1,686	+1,111	+620	-327	-2,223
Classifier + regression	+2,422	+2,025	+1,627	+1,075	+605	-304	-2,122

The pattern that matters is the last column. Every construction that uses the fitted model is under water at 1.00 per leg; both parameter-free constructions are still positive. That is the ordering the main text relies on, and it is why the classifier is an appendix rather than a contribution.

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